

TrustEpisode: Deterministic Episode Formation and Manifest-Scoped Calibration for Partial EDR/NDR Telemetry

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ABSTRACT

Multi-step attacks appear as temporally dispersed endpoint and network observations rather than one decisive alert. Correlation must therefore define which events form a candidate episode, preserve deterministic revisions under late data, and distinguish an unsupported probability from a low probability when telemetry is missing.

We present TRUSTEPISODE, a label-blind EDR/NDR pipeline with explicit ownership boundaries. Evidence Fabric emits canonical events and independent source-health intervals; Episode Synthesis alone owns membership and immutable revisions; a two-feature stateless scorer emits a raw logit; and a manifest-scoped affine calibrator emits a nullable maliciousness probability. A separate deterministic evaluator reports a five-axis evidence-sufficiency profile without changing that probability. An AuditRecord Assembler creates one terminal, append-only outcome per revision: a complete AuditRecord from independently produced inputs, or an explicit AssemblyFailureRecord without fabricated score or probability fields.

We specify a frozen controlled-laboratory protocol for formation, ranking, calibration, partial-telemetry replay, physical collector outage, and deterministic replay. The protocol fixes candidate populations, split assignment, thresholds, group-calibration gates, interventions, matching, metrics, resampling, and empty RT1–RT7 result schemas before locked test access. Empirical result cells remain pending; no synthetic performance values or confidence intervals are reported.

KEYWORDS

attack episode formation, probability calibration, partial telemetry, provenance, EDR, NDR

1 INTRODUCTION

Multi-step attacks rarely produce one conclusive alert. A remote connection, script execution, file transfer, or administrative action can be benign alone yet material when connected by time, entity, session, and process relations. Endpoint detection and response (EDR) and network detection and response (NDR) also fail differently: an event may be absent because no action was observed, because no detector fired, or because a collector or parser was unavailable. Treating all three states as zero evidence silently changes both correlation and probability semantics.

Existing alert correlation and provenance systems reconstruct causal context, but their candidate boundaries, late-data behavior, and calibration contracts are often not explicit enough for byte-identical replay [2, 7–9]. Conversely, an anomaly or severity score is not automatically a probability, and calibration is meaningful only for the candidate population and pipeline version on which it was validated [4, 5]. These gaps are coupled: changing episode membership changes the scored population, while hiding source

failure behind a zero feature can make an unsupported mapping appear confidently benign.

TRUSTEPISODE addresses this problem with a deliberately narrow contract. It transforms only EDR/NDR observations into deterministic EpisodeRevisions, computes a compact raw logit from detector deviation and independent corroboration, emits a calibrated probability only under an exact manifest match, reports evidence sufficiency independently, and assembles one canonical terminal assembly outcome per revision. Ground truth is unavailable to every online producer and is joined only after inference.

Contributions. This paper makes four contributions:

- (1) a single-owner online object model and deterministic episode state machine with fixed guards, total ordering, watermark, and revision policy;
- (2) a compact scorer plus manifest-scoped affine calibration that returns canonical null outside validated support;
- (3) explicit detector-availability and five-axis sufficiency truth tables, together with a schema-ready AuditRecord contract with closed types, nullability, and digest sets; and
- (4) a preregistered controlled-lab protocol with fixed populations, splits, thresholds, perturbations, matching, statistics, and RT1–RT7 result schemas.

2 RELATED WORK

Alert correlation and provenance. Classical alert-correlation methods connect observations into attack scenarios using temporal, causal, and prerequisite relations [9]. Provenance-based intrusion detection preserves process and information-flow context; HOLMES, NODLINK, and KAIROS show how causal graphs support multi-stage reconstruction and investigation [2, 7, 8]. CAPTAIN further demonstrates that a lightweight, interpretable rule system can adapt deployment-specific parameters [14]. TRUSTEPISODE differs by treating deterministic cross-source candidate construction, revision ownership, and candidate-conditioned probability calibration as one versioned contract rather than presenting a new graph detector.

Calibration and incomplete telemetry. Ranking and calibration answer different questions: a high AUPRC does not imply reliable probabilities [4, 5]. Platt-style affine calibration is compact and ranking-preserving when its slope is positive, but only for a fixed candidate population and support regime [11]. Missing-modality research likewise shows that absent inputs require explicit handling rather than silent zero imputation [16]. TRUSTEPISODE therefore separates observed source masks, authenticated health, detector availability, calibration status, and sufficiency. The raw scorer cannot read health or labels, and unsupported records retain a diagnostic logit but no numeric probability.

Scope. The method does not model operational impact, response priority, autonomous response, CTI enrichment, or language-agent behavior. Its claims are limited to deterministic EDR/NDR episode formation, manifest-scoped maliciousness calibration, explicit evidence sufficiency, and replayable audit records.

3 TRUSTEPISODE METHOD

TRUSTEPISODE converts EDR/NDR telemetry into deterministic candidate revisions and version-scoped evidence records. Evidence Fabric and Health Monitor respectively own events and source health; Episode Synthesis alone owns membership; Feature Extractor emits health-blind D, C, m and detector observations; Stateless Scorer emits z ; Scope Resolver emits resolved detector availability, κ , the calibration-group key, and decision_eligible; Calibrator emits nullable P ; and Sufficiency Evaluator emits structured U . For each revision, the AuditRecord Assembler emits exactly one terminal AssemblyOutcome: a canonical AuditRecord when its join succeeds, or an immutable AssemblyFailureRecord when it cannot safely assemble one. Ground truth is joined only after inference. The method estimates malicious activity inside a frozen EDR/NDR candidate; it does not estimate enterprise risk, impact, response priority, or unobservable attacks. Figure 1 shows these boundaries.

3.1 Problem Scope, Output Contract, and Ownership

Let C contain every emitted EpisodeRevision and define

$$C_{\text{sup}} = \{\epsilon^{(r)} \in C : \kappa_{\epsilon^{(r)}} \in \{\kappa_G, \kappa_g\}\}, \quad (1)$$

where κ_G, κ_g , and $\kappa_\emptyset$ serialize as supported_global, supported_group, and out_of_support. The offline target $Y_{\epsilon^{(r)}} \in \{0, 1\}$ is revision specific: evidence first accepted in $r+1$ cannot relabel r . Positive and negative revisions form the primary supervised cohort; mixed and ambiguous revisions enter only the registered sensitivity analysis.

When defined, the candidate-conditioned output is

$$P_{\epsilon^{(r)}} = \Pr(Y_{\epsilon^{(r)}} = 1 \mid \epsilon^{(r)}, v_{\text{episode}}, v_{\text{feature}}, v_{\text{model}}, v_{\text{cal}}). \quad (2)$$

It is not the unconditional probability that an organization is under attack. Online schemas are label blind and reject scenario, split, marker, window, and label fields. Collector identity and health messages are authenticated; activity outside observable EDR/NDR traces is outside the claim.

Table 1 assigns every online object and decision field to one producer and forbids downstream components from silently taking over that ownership.

3.2 Evidence, Health, and Detector Availability

Evidence Fabric maps each source record to a NormalizedEvent with stable ID, event and ingest times, source instance, canonical subject/object/action, immutable raw pointer, and root dependency group. Forwarded copies retain their root and cannot add evidence. OCSF v1.8.0 is only a field-alignment target [10]. Health Monitor independently emits authenticated SourceCoverage intervals from heartbeat, lag, parser, and transport observations; silence is never interpreted as failure. Each manifest fixes a 30 s heartbeat

period: healthy requires no missed heartbeat, lag ≤ 120 s, valid parser, and live transport; degraded means one missed heartbeat or $120 < \text{lag} \leq 600$ s while transport remains live; down requires an authenticated collector/transport stop or two consecutive missed heartbeats; missing or conflicting intervals yield unknown.

For $s \in \mathcal{S}^* = \{\text{EDR}, \text{NDR}\}$,

$$m_{\epsilon, s} = 1[\epsilon \text{ contains an observed event from } s]. \quad (3)$$

Feature Extractor is the sole producer of m and health-independent observation $o_{\epsilon, s}^{\text{det}} \in \{\text{detected}, \text{examined_none}, \text{not_examined}\}$. It reads EpisodeRevision and detector outputs, but never SourceCoverage. Scope Resolver is the sole producer of resolved availability a^{det} and applies, in order: missing or conflicting covering intervals $\mapsto$ unknown; unexpected/down source $\mapsto$ collector_unavailable; invalid parser $\mapsto$ parser_invalid; unknown health or not_examined $\mapsto$ unknown; otherwise detected $\mapsto$ available and examined_none $\mapsto$ no_detection. Thus event silence never becomes health. Finally, $q_{\epsilon, s} = 1$ requires lineage-deduplicated detector support tied to a core entity or edge. Scope Resolver consumes (m, o^{det}, h) , eligible-detector count n , and exact versions and alone emits SupportDecision. It contains $a^{\text{det}}, \kappa, g, e, n$, an eligibility reason, coverage_ref_ids, and health_snapshot_digest; e is decision_eligible. Before calibration resolution, $n = 0$ deterministically yields $\kappa = \kappa_\emptyset$, $e = 0$, reason no_eligible_detector_value, and therefore JSON-null P , even when both resolved states are no_detection. Otherwise $e = 1$ iff all support predicates pass and $\kappa \in \{\kappa_G, \kappa_g\}$. Coverage IDs identify the exact EDR/NDR SourceCoverage objects used by the resolver. The digest covers their RFC 8785 canonical health snapshot.

3.3 Deterministic Episode Synthesis and Revisions

For event e_i and open episode E ,

$$\ell(e_i, E) = 1[\Delta_t \leq 300\text{s} \wedge R = 1 \wedge \text{span}(E \cup \{e_i\}) \leq 3600\text{s} \wedge H = 0]. \quad (4)$$

R comprises process-causal, authenticated-session, endpoint-mapped connection, and specific-entity relations. Their precedence ρ is respectively 4, 3, 2, 1; n_R is the number of distinct satisfied relation classes. $H = 1$ when all support comes from an entity with at least 25 distinct relations in the preceding 600 s. Eligible candidates use the sole order $(-\rho, -n_R, \Delta_t, t_{\text{start}}, \text{episode_id})$.

Zero/one candidates start/attach. Multiple candidates merge only when there are at most three parents, every parent is connected through the new event by a causal, session, or endpoint relation, and the result has at most 500 events and 200 entities; otherwise the first ordered candidate receives the event. Start and merge IDs hash the policy version and canonical inputs; membership order is $(t_{\text{event}}, t_{\text{ingest}}, \text{source_instance}, \text{event_id})$.

The watermark is maximum observed ingest time minus 120 s. When an open lineage first satisfies the watermark, Episode Synthesis emits a finalized successor and freezes t_f and deadline $t_f + 900$ s. The lineage leaves the open-candidate index but remains in a separate revisable index. A closed-lineage index retains canonical relation keys and the latest revision until deadline plus $W_{\text{ret}} = 86400$ s. Before that fixed deadline, a late

Table 1: Single-owner online object contract.

Object	Sole producer	Consumers	Forbidden behavior
NormalizedEvent SourceCoverage	Evidence Fabric Health Monitor	Episode Synthesis Scope Resolver; Sufficiency Evaluator; Assembler	label, split, scenario, or health fields infer outage from event silence; independently resample health
EpisodeRevision / LateEvidenceRecord	Episode Synthesis	Feature Extractor; Sufficiency; Assembler / late store	rewrite a revision; admit beyond-horizon evidence
FeatureVector $[D, C, m, o^{det}]$ Raw logit z SupportDecision Nullable P Structured U AssemblyOutcome	Feature Extractor Stateless Scorer Scope Resolver Calibrator Sufficiency Evaluator AuditRecord Assembler	Scorer; Scope/Sufficiency; Assembler Calibrator; Assembler Calibrator; Sufficiency; Assembler Assembler; downstream review Assembler; downstream review Append-only store; replay; evaluation	SourceCoverage, inferred health, or labels probability wording test-selected status; second eligibility/health producer modification by U ; numeric unsupported output scalarization or recomputation of P both success and failure for one revision; update/delete

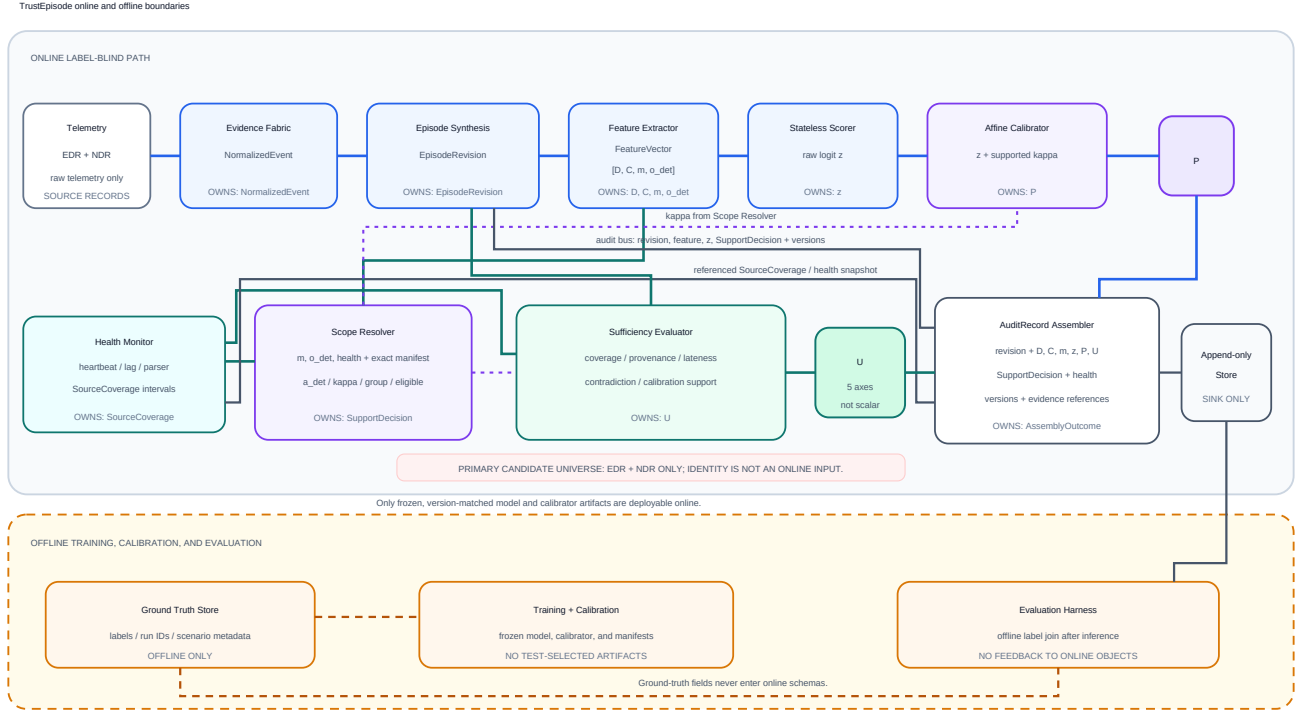


Figure 1: Label-blind online and offline boundaries. The primary candidate path contains EDR/NDR only. Every arrow enters a declared producer; the AuditRecord Assembler owns the terminal AssemblyOutcome and the append-only store is only a sink. Identity is not an online input, and U has no numerical path into the scorer or calibrator.

event is accepted only if Equation 4 holds against the latest revision; acceptance emits a new finalized successor with the unioned membership and the same t_f and deadline. It does not reopen the lineage or extend the horizon. At the deadline a membership-identical closed successor is emitted, removed from the revisable index, and retained in the closed-lineage index. A later related event emits an immutable LateEvidenceRecord with reason `beyond_revision_horizon`, `closed_lineage_match`, or `duplicate_after_terminal_outcome` to the late store; it cannot create an EpisodeRevision or change membership, z , P , or an earlier AuditRecord. An event older than W_{ret} is rejected to RT7 as `closed_index_expired`; it cannot start a fresh episode.

Algorithm 1: label-blind update. Validate and root-deduplicate the event. First query revisable and closed-lineage indexes using

event time and canonical relation keys. If any match, select one by *(state priority, $-\rho, -n_R, \Delta_t, t_{start}, id$)* with revisable before closed; emit exactly one accepted successor or LateEvidenceRecord and return. Only when no non-open lineage matches may the event query sorted open candidates and execute exactly one start/attach/merge action. Events past W_{ret} fail ingress. Advance the watermark and emit each immutable state successor or side record. Replay asserts equality of IDs, membership, links, states, late records, features, outputs, evidence references, and canonical digests. Figure 2 illustrates the fixed transition points.

3.4 Compact Features and Stateless Raw Score

The primary scorer input is

$$x_\epsilon = [D_\epsilon, C_\epsilon]. \quad (5)$$

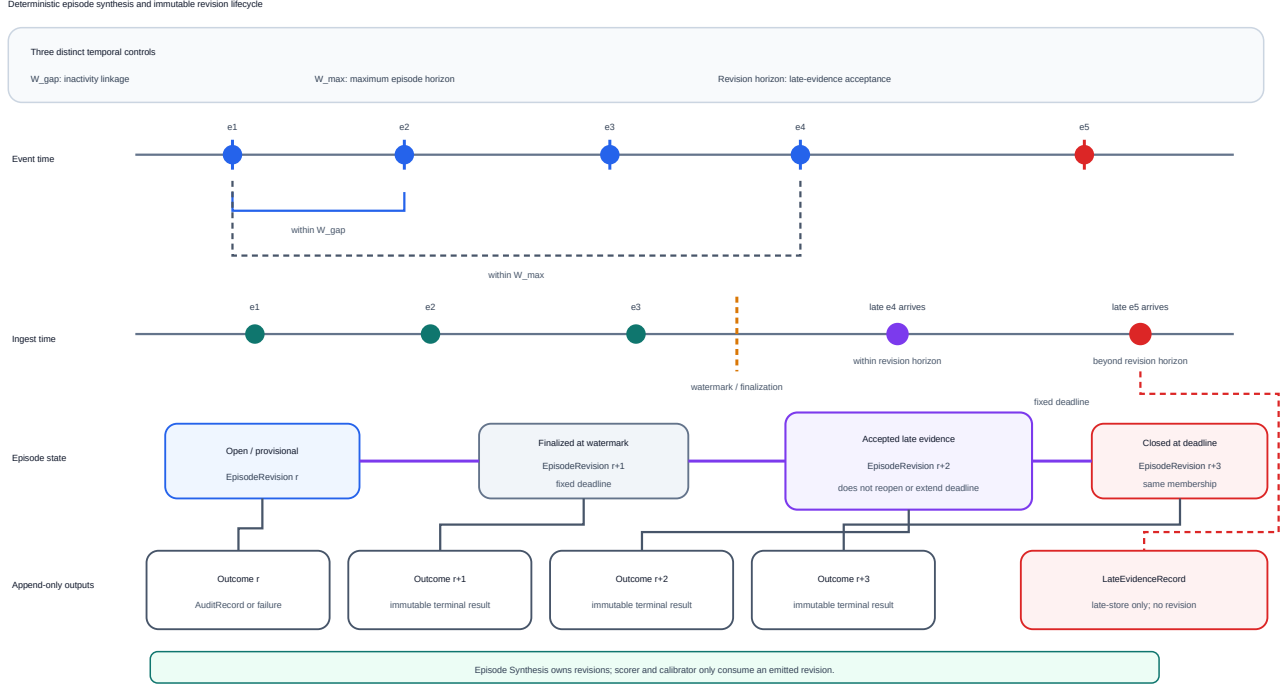


Figure 2: Deterministic lifecycle. Event time controls linking, ingest time controls the 120 s watermark, accepted evidence within 900 s creates a successor revision, and evidence beyond the horizon cannot rewrite history.

The frozen detector registry contains exactly three EDR IDs and three NDR IDs; each emits only a finite binary64 value $r \in [0, 1]$, with larger values more suspicious. Source family, detector ID, and dependency group form the deduplication key. Byte-identical retries collapse; a conflicting value or reference rejects the FeatureVector. For each detector, negative training revisions define $\hat{F}(r) = (\#\{v < r\} + 0.5\#\{v = r\})/N$; $N = 0$ invalidates the registry. Sort normalized values descending, breaking ties by UTF-8 detector ID then dependency-group ID. If $n > 0$, D is the mean of the first $\min(3, n)$ values; if $n = 0$, $D = 0$ is a diagnostic placeholder and the revision is not decision eligible. Missing/nonfinite/out-of-range values are never zero-imputed. For source s , $q_{\epsilon, s} = 1$ iff at least one root-deduplicated registered detector has detected, a legal value, and a root tied to a core entity or selected edge; otherwise it is zero. Thus corroboration is fixed to

$$C_{\epsilon} = q_{\epsilon, \text{EDR}} q_{\epsilon, \text{NDR}}, \quad K = 2. \quad (6)$$

Identity is only an offline M5-ID annotation and cannot alter candidates or features.

With $\lambda = 10^{-4}$, the stateless scorer is the unique frozen solution

$$\begin{aligned} (\hat{b}, \hat{w}_D, \hat{w}_C) = \arg \min_{b \in \mathbb{R}, w_D, w_C \geq 0} & \frac{1}{N} \sum_i [\log(1 + e^{z_i}) - y_i z_i] \\ & + \frac{\lambda}{2} (b^2 + w_D^2 + w_C^2), \end{aligned} \quad (7)$$

$$z_i = b + w_D D_i + w_C C_i.$$

Its cohort is the first finalized revision per training lineage with exact EDR/NDR health/parser support and an offline positive/negative label; mixed, ambiguous, OOS, and failed outcomes are excluded and counted. Deterministic binary64 L-BFGS-B starts at zero, uses stable $\logaddexp$, stops at projected-gradient infinity norm $\leq 10^{-9}$ and objective change $\leq 10^{-12}$, and fails closed after 10,000 iterations. The RFC 8785/SHA-256 model artifact covers ordered training revision IDs/labels, detector-registry and normalizer digests, matrix bytes, λ , solver contract, and final coefficients. Health, m , a^{det} , κ , U , and outage status never enter x_{ϵ} .

3.5 Manifest-Scoped Affine Calibration

The global calibrator uniquely minimizes natural-prevalence calibration NLL plus $10^{-6}(a^2 + b_{cal}^2)$ subject to $a \geq 10^{-6}$. Thus

$$P_{\epsilon} = \begin{cases} \sigma(az_{\epsilon} + b_{cal}) & e = 1, \kappa = \kappa_G, \\ \sigma(a_g z_{\epsilon} + b_g) & e = 1, \kappa = \kappa_g, \\ \text{null} & e = 0, \kappa = \kappa_{\emptyset}, \end{cases} \quad (8)$$

following positive-slope score calibration [4, 11]. The primary manifest supports only exact EDR+NDR mask 11, non-down/non-unknown health, valid parsers, resolved detector states available or no_detection, and exact pipeline versions. Any mismatch is out-of-support.

Before calibration, the manifest registers each group key as the RFC 8785/SHA-256 digest of a canonical object containing deployment profile and EDR/NDR detector versions; scenario, run, identity, and label fields are forbidden. Conditional groups

use deterministic balanced five-fold run-level cross-fitting on the calibration partition. Within each group, intact runs with binary counts (p_r, n_r) sort by decreasing $p_r + n_r$, then seeded SHA-256 and run ID. Each run goes to the fold minimizing the resulting maximum normalized positive, negative, and run loads; ties use total normalized load then fold ID. Every fold must be nonempty, and the ordered assignment map is sealed by digest. A group requires at least 50 positive revisions, 50 negatives, and 10 runs. For each held-out fold, improvement is $\Delta B = \text{Brier}_{\text{global}} - \text{Brier}_{\text{group}}$ and $\Delta N = \text{NLL}_{\text{global}} - \text{NLL}_{\text{group}}$, first macro-averaged by run and then equally across folds. Both means must be at least 0.01, every fold must be nonnegative, and paired run-bootstrap lower 95% bounds for both must exceed zero. A passing group is refit once on all calibration-partition records with that key; its final coefficients and key are sealed in the manifest. Manifest validation rejects duplicate keys. Runtime resolution is exclusive: first reject any support mismatch; otherwise select the single enabled exact key, or fall back to global when none is enabled. Scope Resolver alone emits g, κ , and $e = 1[n > 0 \wedge \kappa \neq \kappa_\emptyset]$; Calibrator only maps (z, e, κ, g) to P . This invariant prevents no_detection health support from making an empty detector vector score eligible.

3.6 Structured Sufficiency and Audit Assembly

$$U_\epsilon = (u_{\text{coverage}}, u_{\text{provenance}}, u_{\text{lateness}}, u_{\text{contradiction}}, u_{\text{calibration}}). \quad (9)$$

Table 2 makes every U axis a first-match deterministic rule rather than an informal confidence judgment.

The frozen contradiction rules are CR1, overlapping authenticated endpoint bindings map one endpoint key to different hosts; CR2, matched EDR socket and NDR flow disagree on the initiator after canonical direction normalization; and CR3, the same $(\text{host}, \text{boot}, \text{pid}, \text{start})$ process key has conflicting parent/root lineage. A rule is applicable only when its relation exists; absent required fields then yield unevaluable. No applicable relation yields none. U is never scalarized and cannot gate, modify, or relabel P .

AuditRecord Assembler joins one EpisodeRevision, FeatureVector, z , SupportDecision, P , U , and the exact SourceCoverage objects referenced by SupportDecision, by $(\text{episode_id}, \text{revision})$ and exact versions. It copies the health snapshot from those references; it cannot independently select or infer health. Required provenance comprises every membership event/root, selected link, scored input, coverage object, and version used by z, P, U ; only unscored diagnostic references are optional. Identical retries are idempotent; conflicting duplicates fail. At deterministic deadline $\text{revision.watermark} + 120$ s it atomically inserts exactly one AssemblyOutcome under that key: a complete AuditRecord, or an AssemblyFailureRecord with one code from $\{\text{assembly_timeout}, \text{missing_input}, \text{conflicting_duplicate}, \text{key_mismatch}, \text{version_mismatch}, \text{schema_invalid}, \text{digest_mismatch}\}$ and no fabricated score/probability fields. The unique-key store grants insert/select only and forbids a later success after terminal failure. RT1/RT7 count failures. Inputs are processed in fixed type order EpisodeRevision, FeatureVector, raw logit, SupportDecision, P , U , SourceCoverage. When multiple failures apply, precedence is schema, key, duplicate, digest, version, missing-input, then

timeout. Before canonicalization, references sort by relation rank (membership/link/feature/health/contradiction), dependency-group UTF-8 bytes, object reference before sentinel, object-ID-or-empty bytes, then digest-or-empty bytes. RFC 8785 preserves JSON array element order rather than sorting it; JCS is applied only after this application-level ordering. Optional absence uses a closed sentinel; missing required provenance fails assembly. Every closed VersionSet component binds a version identifier and SHA-256 digest for schema/parser/formation, detector registry, feature, normalizer, raw model, calibrator, support, sufficiency, assembly, and manifest. JSON numbers must be finite binary64 values represented by JSON decimal grammar; NaN, infinities, negative zero, and overflow fail schema validation. Runtime host, PID, wall-clock, and duration form a non-canonical envelope excluded from digests.

3.7 Boundary and Reproducibility Contract

The method excludes CTI, RAG/agents, impact/priority, automated response, Identity as an online source, and OT/ICS generalization. Appendix A.1 and the tables cited in the main text are normative. The required supplementary TrustEpisode_reproducibility_companion_v1.zip contains the executable JSON Schema and registries; its SHA-256 is 6298bd2e89bd50a78a890b62750dcca2bcee34906fd6dd73a4a77f2a19c81205. A missing or mismatched upload makes the machine-contract claim unverified. The contribution is limited to deterministic EDR/NDR episode construction, manifest-scoped probability, structured sufficiency, and replayable append-only provenance.

4 EVALUATION

The evaluation is a frozen protocol, not a report of completed experiments. Populations, splits, manifests, thresholds, gates, interventions, matching, metrics, resampling, and empty RT1–RT7 schemas are fixed before locked-test access. Every unexecuted cell is pending; no synthetic value, curve, or interval is reported.

4.1 Research Questions and Populations

RQ1 – formation. Does the label-blind builder increase malicious-action coverage over time-only and time-plus-entity comparators while reducing fragmentation and over-merge? **RQ2 – ranking/calibration.** Does $D + C$ improve AUPRC and fixed-budget recall over severity and D -only comparators, and does independent affine calibration improve Brier, NLL, and reliability? **RQ3 – partial telemetry.** Under paired replay and separate physical outages, how do formation, z , support, U , revisions, and eligibility change? **RQ4 – replay/cost.** Are canonical outputs byte identical, and what throughput, latency, memory, storage, and replay time are measured?

The populations are disjoint views of the same registered runs. C_{form} contains one terminal revision per surviving episode lineage at $T_{\text{form}} = t_{\text{run_end}} + L_{\text{wm}} + W_{\text{rev}} = t_{\text{run_end}} + 1020$ s; merged parents are removed. $C_{\text{dec}}^{\text{all}}$ contains the AssemblyOutcome of the first finalized revision per terminal lineage, including supported/OOS AuditRecord and AssemblyFailureRecord. The

Table 2: Deterministic U rules; rows list highest-to-lowest precedence.

Axis	Precedence	First matching predicate
coverage	unknown > unavailable > degraded > complete	missing/conflicting required interval; else any down/not-collected/parser-invalid; else one missed heartbeat or lag (120, 600] s; else all required intervals healthy
provenance	missing > partial > complete	unresolved scored input, membership event, selected link, coverage interval, or version; else optional unscored diagnostic absent; else all required roots resolve
lateness	late_accepted > revisable > stable	successor accepts late evidence; else open/finalized before fixed deadline; else closed
contradiction	present > unevaluable > none	any evaluable CR1-CR3 true; else an applicable rule lacks a required field; else all applicable rules false
calibration	out_of_support; supported_group; supported_global	exact serialization of Scope Resolver κ

scorable inventory is

$$C_{dec}^{score} = \{o \in C_{dec}^{all} : o \text{ is AuditRecord, } o.decision_eligible, P \in [0, 1]\}.$$

After every terminal outcome is inserted, the offline harness joins label $L(o) \in \{positive, negative, mixed, ambiguous\}$ by its EpisodeRevision digest, including for AssemblyFailureRecord. The binary metric population is

$$C_{dec}^{binary} = \{o \in C_{dec}^{score} : L(o) \in \{positive, negative\}\}.$$

C_{stream} contains every terminal AssemblyOutcome keyed by $(episode_id, revision)$, including AssemblyFailureRecord, and is used only for RQ3/RQ4 transition and cost analysis. Every partition/run reports supported-global, supported-group, OOS-AuditRecord, and failure counts and rates with denominator $|C_{dec}^{all}|$; scorable coverage uses $|C_{dec}^{score}|$, while every class-dependent ranking, discrimination, and proper-score metric uses $|C_{dec}^{binary}|$. RT1 also reports outcome and failure counts by all four offline labels, exposing class-dependent assembly failure. Thus late revisions cannot duplicate RQ1/RQ2 candidates and assembly failure cannot induce hidden survivorship bias.

4.2 Controlled Laboratory and Frozen Split

Formal claims use only a manifest-pinned controlled Linux laboratory; public provenance datasets are related-work context, not an evaluation tier. Its four-service primary topology is Caldera v5.3.0 [1], two isolated Ubuntu 22.04 targets, and one inline Suricata gateway. Each target command passes through an instrumented EDR wrapper that records generic process start/exit observations to an internal append-only spool; an independently stoppable forwarder emits label-free events and authenticated health. This is container instrumentation, not a native commercial endpoint product. A host monotonic orchestrator executes frozen no-profile Bash card templates. Sandcat is disabled during formal evidence windows, retaining the pinned Caldera control-plane topology without admitting agent identifiers or control traffic into primary telemetry. The single NDR adapter must pass canonical-byte, digest, event-ID, direction, and protected-IP conformance against the pinned sensel-ot-edge-sensor implementation. Wazuh, guacamole-ai, and Aristaconnector-Control-Plane may inspect only a post-seal copy and cannot affect RT1-RT7. The Caldera release pin is annotated tag object dd22d543... peeled to commit b24f6e7a...; each run also records image, configuration, topology, parser, schema, source, and hardware digests. Missing fields fail closed into RT1 only.

Table 3 fixes the split unit, deterministic assignment, descendant inheritance, and the fail-closed registry fields used by every result table.

Identity is never an online source. M5 is an EDR/NDR endpoint card; M5-ID is a separately labeled post-inference annotation admitted only after its timestamp/provenance/health/support pilot passes. It cannot alter S^* , $K = 2$, candidates, features, manifests, or primary RQs.

4.3 Ground Truth, Matching, and Labels

Allowlist schemas reject operation_id, ability_id, adversary/technique labels, action receipts/windows, scenario class, malicious label, and split assignment from every online object [6]. Each ground-truth group g registers a nonempty set of required action tokens A_g ; each run registers benign/background tokens B_r . For predicted episode p , $A_{cov}(p, g) = |A_p \cap A_g|/|A_g|$, where A_p is its root-deduplicated attributed action set. Offline matching constructs the threshold-qualified edge graph E_q . An edge first requires temporal IoU ≥ 0.10 , entity Jaccard ≥ 0.50 , and one exact offline action receipt or registered action. Its weight is

$$w = 0.5A_{cov} + 0.3J_{entity} + 0.2IoU_{time}; \quad w \geq \theta_{match} = 0.50. \quad (10)$$

Maximum-weight one-to-one bipartite assignment over E_q operates within one run; ties use entity overlap, absolute time difference, then ascending ASCII/UTF-8 bytes of $(episode_id, group_id)$. Assignment is used only for episode precision/recall/F1. Coverage, fragmentation, and over-merge use unassigned E_q .

A revision chooses the first incident edge under descending (A_{cov}, w) then ascending UTF-8 group ID; with no edge, $g^* = \text{null}$, $A_{cov} = 0$, and B_{cont} is undefined. Otherwise contamination is $B_{cont} = |A_p \cap B_r|/|(A_p \cap A_{g^*}) \cup (A_p \cap B_r)|$; a zero denominator is undefined. First-match label precedence is: mixed if $A_{cov} \geq \theta_{mal} = 0.50$ and $B_{cont} \geq \theta_{contam} = 0.25$; positive if coverage passes and contamination is below the threshold; negative if $A_{cov} = 0$, at least one registered benign token is present, and no conflict exists; otherwise ambiguous, including undefined contamination. A truth conflict exists iff overlapping registered groups assign one root/action token to different group IDs, or the same root is both malicious and benign, or duplicate receipt IDs disagree on card/window; any conflict forces ambiguous before assignment. Labels are joined only after terminal AssemblyOutcome insertion; the join resolves EpisodeRevision independently of whether assembly succeeded, and an unresolved revision is counted as ambiguous. Sensitivity varies $\theta_{mal} \in \{.25, .50, .75\}$, $\theta_{contam} \in \{.10, .25, .50\}$, and $\theta_{match} \in \{.40, .50, .60\}$ without altering the primary registry. Figure 3 shows the enforced join boundary.

Table 3: Frozen campaign split and descendant inheritance.

Item	Unique rule	Fail-closed invariant
Independent unit	base campaign, stratified by scenario family and benign mode	no event/episode random split
Order	ascending SHA-256 of 20260710:base_campaign_id per stratum	freeze IDs before label release
10-run cycle	ranks 0–4 train; 5–6 development; 7 calibration; 8–9 locked test	exact 50/20/10/20 percent; all partitions nonempty
Descendants	replay children inherit partition and bootstrap cluster; dedicated PO1 cells are locked-test-only independent clusters	any mismatch or PO1 reuse in fitting invalidates the complete cluster
Registry	run, base/parent/pair IDs, scenario, benign mode, seed, partition, cluster, role, manifest digest	missing field excludes RT2–RT7

Ground Truth Firewall and offline label join

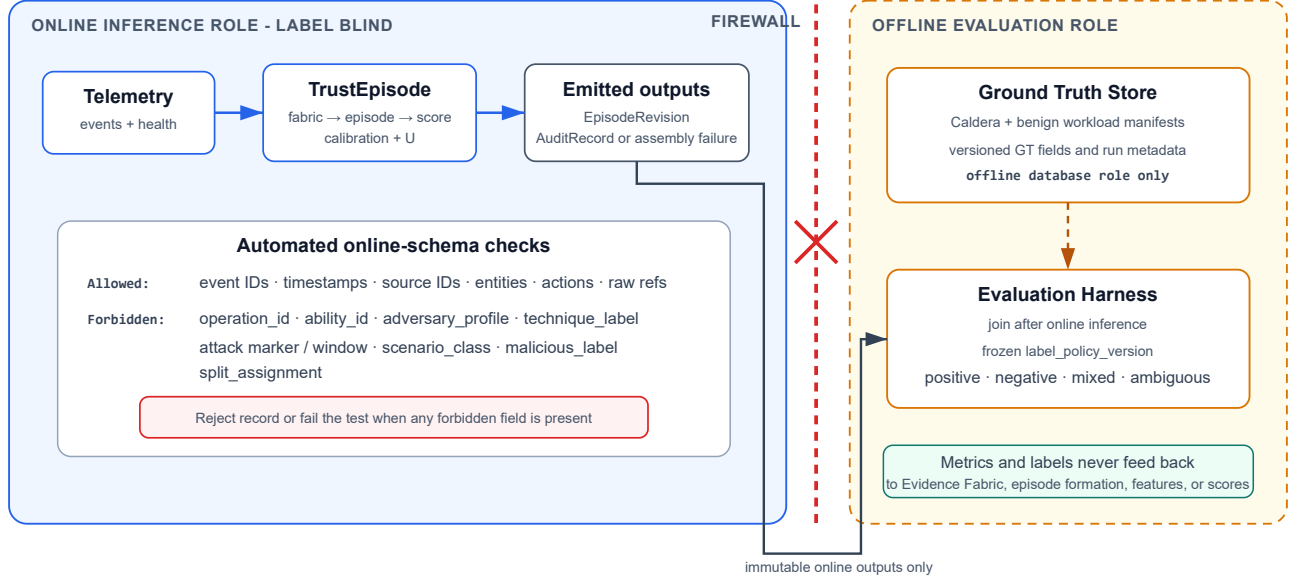


Figure 3: Ground Truth Firewall. Schema tests reject evaluation-only fields before inference; labels join immutable outputs only inside the offline Evaluation Harness.

4.4 Scenario and Benign Cards

Table 4 defines the seven malicious families; it specifies required execution counts, not achieved observations.

B1–B8 respectively cover Linux inventory, manifest-pinned package installation, file synchronization, registered service scanning, read-only SSH administration, log collection, configuration management, and supervisor-backed maintenance. The protocol requires 10 valid pure-benign repetitions per card; achieved counts remain pending. Every card fixes preconditions, fixed-path post-conditions, offline-only action receipts, expected observations, cleanup, and card-specific failure. Collection failure remains in RT1/failure logs but leaves RT2–RT7; action failure with healthy collectors enters only the sensitivity cohort; cleanup failure blocks image reuse. This paragraph and Table 4 are the normative card contract for this PDF.

Each M1–M7/B1–B8 card is additionally a closed execution tuple: image-manifest key; ordered ability IDs bound to a no-profile Bash executor and exact command template; monotonic start offsets; fixed target paths; offline receipt SHA-256 preimage; success and failure predicates; cleanup commands and post-cleanup digest check; and allowed mode. Target commands cannot receive

run/card/ability/operation/receipt IDs or their values. The malicious modes are attack-only and one deterministic concurrent B1–B8 card starting 60 s earlier; benign cards also run standalone. Unresolved template variables, command/image digest mismatch, offset error > 1 s, or cleanup failure invalidates the run. These tuples are frozen before formal collection.

4.5 Calibration Protocol and Comparators

Training fits the raw scorer; development freezes the contract; calibration fits the global mapping and executes the Section 3.5 five-fold group gate; locked test selects nothing. Supported results report counts, prevalence, Brier/NLL, ten equal-width reliability bins with counts, and run-cluster intervals. Unsupported records report coverage and status but receive no proper score. Model fitting, calibration, AUPRC/AUROC, Brier, NLL, and reliability use C_{dec}^{binary} only. The global positive-slope mapping must preserve AUPRC/AUROC on that same binary population or trigger an implementation audit.

Table 5 binds every formation, ranking, calibration, replay, and outage comparator to one implementation and one RT registry.

Table 4: Frozen malicious families; the protocol requires 10 valid repetitions per card in each mode; achieved counts are pending.

ID	Ordered family	Stressor	Primary evidence
M1	discovery → peer inventory → SSH write	lateral chain	EDR lineage + NDR connection
M2	stage → archive → SCP → delayed local action	weak-to-strong	EDR file/process + NDR transfer
M3	discovery → temporary-key SSH write	remote execution	Linux EDR + NDR session
M4	five-request 60 s cadence → local action	sparse sequence	NDR cadence + EDR action
M5	discovery → local configuration action	source-asymmetric hard case	EDR; valid NDR no-detection
M6	matched-tool SSH → remote write	tool-name shortcut	contextual EDR/NDR
M7	actions at 0/300/3900 s	300/3600 s boundaries	fragmentation/stabilization

Table 5: Registered comparisons; each ID has one implementation and one result location.

ID	Scope	Single implementation	Purpose / registry
EB1	formation	non-overlapping 300 s event-time windows	time-only / RT2
EB2	formation	EB1 plus one specific shared entity	time-entity / RT2
EB3	formation	proposed guarded temporal-causal builder	reference / RT2
EB3-H	formation	EB3 with hub suppression disabled	component ablation / RT2
SB1	ranking	maximum normalized source-native severity	external rank / RT3a
SB2	ranking	constrained D -only logit	corroboration ablation / RT3a
SB3	ranking	proposed $D + C$ logit	raw rank / RT3a
SB4	score	uncalibrated sigmoid score $\sigma(z)$	non-probability comparator / RT3b
SB5	calibration	global affine mapping	primary / RT3b
SB6	calibration	enabled registered group mapping	conditional / RT3b
SB7	replay	unavailable detector forced to zero; support ignored	anti-pattern / RT4
OB1	outage	available-source denominator; < 2 healthy gives no corroboration	outage only / RT5

The baseline registry closes the abbreviated table rows. EB1 assigns admitted events to $\lfloor (t^{event} - t_{run_start})/300s \rfloor$ and emits one episode per nonempty bin. EB2 forms exact shared-entity connected components inside each EB1 bin, with entity-less events as singletons. SB1 is the maximum root-deduplicated finite source_native_severity $\in [0, 1]$ produced by the pinned parser; missing values are undefined and counted. SB2 fits the same binary training cohort as SB3 using $b \in \mathbb{R}$, $w_D \geq 0$, $\lambda = 10^{-4}$, and the same deterministic L-BFGS-B tolerances, then ranks by $b + w_D D$. Canonical IDs, tie orders, failure handling, and artifact digest fields are fixed by the baseline registry.

4.6 Replay Perturbations and Physical Outage

The operators define deterministic children for every M1–M7 base run, but formal RT4 instantiates only the 28 malicious locked-test roots (seven cards, two modes, two locked repetitions). Each root has 41 RP0–RP7 children, for 1,148 sealed replay bundles; non-locked roots are never read by RT4. RP2–RP7 instantiate the Cartesian product of source $\in \{\text{EDR}, \text{NDR}\}$ and registered dose; RP1 explicitly uses EDR-only/NDR-only and RP0 has no source. Let

$$\begin{aligned}
 M &= \text{base} \parallel 0x1f \parallel \text{RP} \parallel 0x1f \parallel \text{source} \\
 &\quad \parallel 0x1f \parallel \text{JCS}(\text{level}), \\
 S &= \text{HMAC}_{\text{SHA256}}(20260710, M), \\
 U(d, k) &= \text{uint64be}(\text{HMAC}_{\text{SHA256}}(S, d \parallel 0x1f \parallel k)[0:8]) \\
 &\quad / 2^{64}.
 \end{aligned}$$

Base events sort by $(t^{event}, t^{ingest}, \text{source_instance}, \text{event_id})$. RP0 is unchanged; RP1 masks one family; RP2 drops explicit-source events at 5/10/25%; RP3 removes an explicit-source 30/120/300 s half-open interval; RP4 delays ingest by 30/120/600 s; RP5 reorders within 30/120/600 s buffers anchored at minimum base ingest time; RP6 shifts event time by $-120/-30/+30/+120$ s; and RP7 duplicates explicit-source root groups at 5/10/25%. Other-source events and SourceCoverage remain unchanged.

RP3 uses $j = \lfloor U(\text{interval}, \text{start})N \rfloor$ over N ascending integer-microsecond valid starts; $N = 0$ invalidates the child. RP5 width is exactly the dose in integer microseconds and its within-buffer key is $(U(\text{permute}, \text{source} \parallel \text{event_id}), \text{event_id})$. RP7 preserves the root and sets the duplicate ID to SHA-256 of byte preimage the concatenation of UTF-8 RP7, separator $0x1f$, raw S_{32} , another separator, UTF-8 event ID, a final separator, and uint32be(1), with ingest time increased by one microsecond; collision invalidates the child.

PO1 is a separate locked-test-only cohort, not a child of all 220 base runs. It expands $\{\text{EDR}, \text{NDR}\} \times \{120, 600\} \times 10$ repetitions into exactly 40 physical traces. With card order M1/M3/M4/M7, source index EDR/NDR = 0/1, duration index 120/600 = 0/1, and repetition $r \in [0, 9]$, the card index is $(r + 2i_{\text{source}} + i_{\text{duration}}) \bmod 4$, assigning exactly 10 traces per card. TrustEpisode and OB1 reuse the same immutable trace and health journal. For each cell, t_{anchor} begins after two healthy heartbeats with valid parser, live transport, and backlog ≤ 120 s. Continuous health for 120 s gives $t_{\text{stop}} = t_{\text{anchor}} + 120$ s. After authenticated collector stop, the allocated card begins and at least one ability executes during the outage; the collector remains stopped until $t_{\text{restore}} = t_{\text{stop}} + d$. The registry records t_{detect} , t_{h1} , t_{h2} , and first t_{ready} beginning two consecutive parser-valid, live, backlog- ≤ 120 heartbeat intervals; $t_{\text{stable}} \geq \max(t_{h2}, t_{\text{ready}})$ is followed by 900 s without a revision. Detection is right-censored at restore if absent; recovery is right-censored at $t_{\text{censor}} = t_{\text{restore}} + 1800$ s if ready/stable is absent. Uncensored durations are $t_{\text{detect}} - t_{\text{stop}}$ and $t_{\text{stable}} - t_{\text{restore}}$. Each PO1 trace is its own bootstrap cluster and is excluded from fitting and base-run RT2/RT3 populations. Figure 4 summarizes the non-pooled designs.

4.7 Metrics and Run-Level Uncertainty

For run r , let $E_{q,r}$ be its unassigned threshold-qualified graph. For each ground-truth group g , $n_g = |\{p : (p, g) \in E_{q,r}\}|$; for episode p , $d_p = |\{g : (p, g) \in E_{q,r}\}|$. RQ1 uses run-macro action coverage, $|G_r|^{-1} \sum_g \max(0, n_g - 1)$ fragmentation, and $|P_r|^{-1} \sum_p 1[d_p > 1]$

Paired partial-telemetry perturbations and separate physical-outage validation

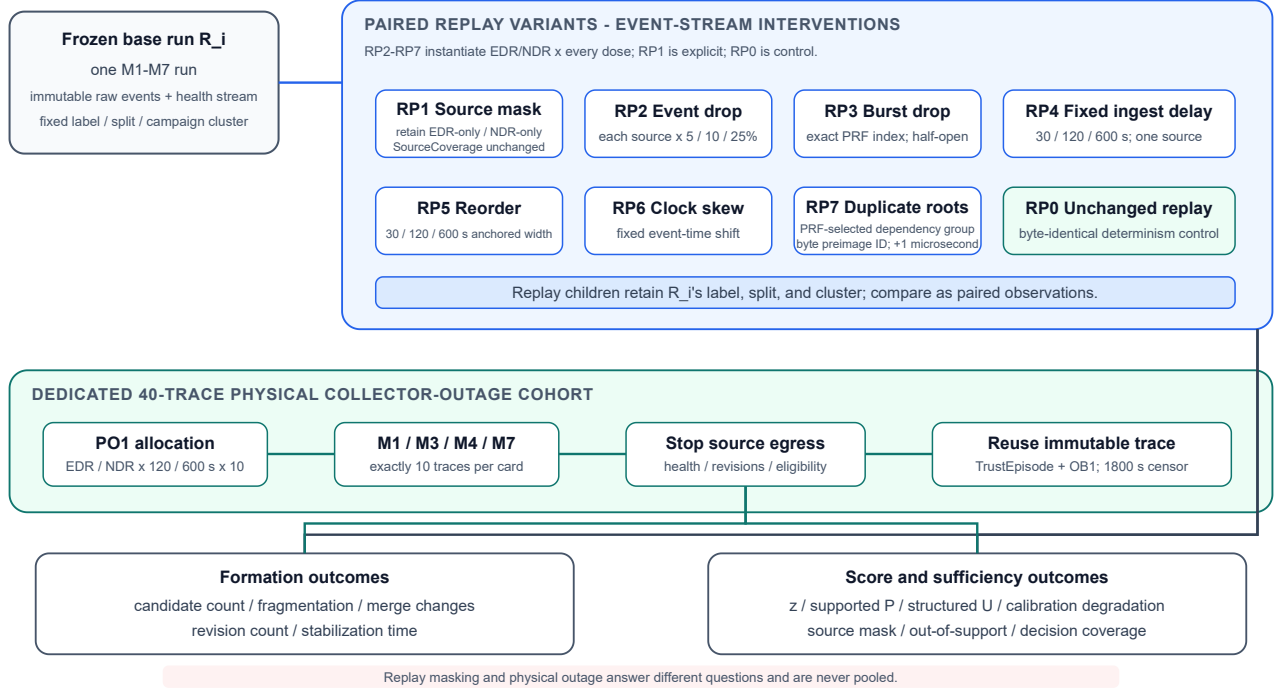


Figure 4: Paired replay changes events with fixed health; the separate 40-trace PO1 cohort changes authenticated health and reuses each trace for both outage comparators.

over-merge. Run action coverage is $|G_r|^{-1} \sum_g \max_p A_{cov}(p, g)$; run contamination is the arithmetic mean over defined B_{cont} values in $C_{form,r}$, with undefined count reported separately. Empty G_r or P_r makes the corresponding metric undefined, never zero. One-to-one assignment is used only for episode F1, so neither graph metric is forced to zero.

RQ2 uses C_{dec}^{binary} only: AUPRC and recall among the top $\max(1, \lceil 0.10 |C_{dec,r}^{binary}| \rceil)$ per run are primary ranking metrics; Brier, NLL, and ten-bin reliability use that same positive/negative population only. AUROC and precision@ k are secondary [13]. RQ3 reports paired run-macro changes and all U transitions; RT5 reports outage detection/recovery. RQ4 checks byte equality and measured cost. RT6 replays frozen RP0 inputs at worker counts 1/2/4/8 on the manifest-pinned host, with one excluded warm-up and five measured repeats per run/worker. A monotonic clock measures ingest-to-AssemblyOutcome latency; RSS is sampled every 100 ms; storage sums canonical bytes; throughput includes all terminal outcomes, and each row records hardware/software manifest digest and failure fraction.

All primary summaries are macro-averaged over runs; pooled micro results are supplemental. Intervals use 2,000 percentile bootstrap resamples of complete run clusters with seed 20260710 [3]; replay/outage resample base-run pairs. Undefined class-dependent resamples are counted and omitted, never zeroed. Relative effects are supplemental.

4.8 RT1-RT7 Reporting Contract

Table 6 is the sole normative result registry; no external machine registry is assumed available to reviewers.

Planned reliability and revision plots remain absent until locked outputs exist. Any later machine-readable registry must exactly preserve the table's RT keys, rows, columns, pending, and N/A rules.

4.9 Limitations and Safety

The controlled lab cannot establish field generalization or every evasion/base rate. Calibration applies only to exact supported manifests. Identity remains offline. Formal runs use isolated hosts, test accounts, offline-only action receipts, cleanup gates, and recorded failures. The evaluation does not study CTI, agents, impact, priority, OT/ICS, or autonomous response, and reports no empirical result until the locked pipeline executes.

5 CONCLUSION

TRUSTEPISODE defines a narrow, replayable contract for EDR/NDR episode formation and candidate-conditioned maliciousness calibration. Episode Synthesis alone owns immutable membership revisions; the scorer, support resolver, calibrator, sufficiency evaluator, and AuditRecord Assembler have disjoint outputs; each revision terminates in either a complete AuditRecord or an explicit AssemblyFailureRecord; and missing or incompatible telemetry yields an explicit out-of-support state rather than a fabricated

Table 6: Frozen result registries. Every unexecuted field serializes as pending; only structurally inapplicable fields use N/A.

Registry	Rows	Frozen columns	State
RT1	partition $\times$ population	runs; $Call/Cscore/C^{binary}$; supported-global/group, OOS, failure counts/rates; outcome/failure labels; prevalence	pending
RT2	EB1, EB2, EB3, EB3-H	action coverage; fragmentation; over-merge; contamination; episode F1	pending
RT3a	SB1, SB2, SB3	all/score/binary denominators; score/binary coverage; AUPRC; recall@budget; AUROC; precision@k	pending
RT3b	SB4, SB5, SB6	all/score/binary denominators; Brier; NLL; reliability bins; supported/OOS/failure rates	pending
RT4	RP0-RP7 $\times$ source $\times$ level	formation/z/Brier/NLL changes; coverage; U transitions; revisions; stabilization	pending
RT5	source $\times$ duration $\times$ method/OB1	health detection; eligibility; OOS; recovery; stable revision time	pending
RT6	worker count 1/2/4/8	manifest; repeats; hash equality; throughput; p50/p95/p99; peak RSS; canonical bytes/event/outcome; replay time; failure fraction	pending
RT7	error category	count; run/candidate fraction; representative revision; adjudication	pending

probability. The accompanying protocol fixes implementation and evaluation decisions before locked-test access. Empirical results remain pending, so the present contribution is the deterministic method and preregistered evaluation contract, not a claim of measured superiority or deployment generalization.

A FROZEN SUBMISSION CONTRACTS

A.1 Machine Schema and Audit Assembly

The normative schema-ready in-PDF contract follows JSON Schema Draft 2020-12 closed-object semantics [15] for NormalizedEvent, SourceCoverage, EpisodeRevision, LateEvidenceRecord, FeatureVector, SupportDecision, AuditRecord, and AssemblyFailureRecord. FeatureVector owns D, C, m, o^{det} without health input. SupportDecision owns resolved a^{det}, κ, g , detector count n , eligibility reason, decision_eligible, EDR/NDR coverage_ref_ids, and health_snapshot_digest. Count $n = 0$ forces out-of-support, false eligibility, and null P even if both detectors report no_detection. AuditRecord requires those outputs, z, P, U , the revision digest, health snapshot, ordered references, exact versions, and canonical digest; out-of-support requires JSON null P and false eligibility. For each revision key the Assembler atomically inserts one terminal outcome: AuditRecord after a complete exact-version join, or AssemblyFailureRecord after conflict or deadline. The assembly deadline is revision.watermark + 120 s. The unique-key store is an insert/select sink and cannot later replace failure with success.

Identifiers are ASCII and sort by unsigned UTF-8 bytes. AuditRecord uses string IDs/digests, nonnegative integer revision, finite $D, P \in [0, 1]$, $C \in \{0, 1\}$, numeric z , Boolean eligibility, closed enums/maps for κ, U, m, h, a^{det} , ordered reference arrays, and a closed version object. Only P and group key are nullable: out-of-support requires $P = \text{JSON null}$; only supported-group requires a non-null key. NaN, infinities, negative zero, and binary64 overflow are invalid. AssemblyFailureRecord contains no D, C, z, P, U and has exactly one of the seven failure codes in Section 3.6. Canonical bytes follow RFC 8785 JSON canonicalization [12]. JCS sorts object properties but preserves array order, so the application sorts every reference array first. SHA-256 includes each outcome’s declared digest-field set except canonical_digest. AuditRecord includes type/schema/key/revision, revision digest, $D, C, z, P, \kappa, g, e, U, m, h, o^{det}, a^{det}$, detector count/reason, ordered references, and versions; AssemblyFailureRecord includes type/schema/key/revision, revision digest, failure code, observed input types, and assembly-policy version; LateEvidenceRecord includes type/schema/episode/latest revision and digest/event,

arrival/deadline/reason, raw pointer/root, ordered lineage references, and VersionSet. Host, PID, wall time, and duration are an optional non-canonical envelope. References sort by relation rank, dependency-group bytes, object-before-sentinel, object-ID-or-empty bytes, then digest-or-empty bytes. Optional absence uses a closed sentinel; required absence fails assembly. Assembly input order is revision, feature, logit, support, probability, U , then coverage. Failure precedence is schema, key, duplicate, digest, version, missing, then timeout. Every VersionSet component closes both a version identifier and content digest for schema, parser, formation, detector registry, feature, normalizer, model, calibrator, support, sufficiency, assembly, and manifest. Conforming validation must accept supported, unsupported, late, and assembly-failure records and reject any online record containing malicious_label. Online objects reject all label, scenario, marker/window, ability/operation, and split fields.

A.2 Episode Algorithm and State Machine

Fixed values are $W_{gap} = 300$ s, $W_{max} = 3600$ s, $L_{wm} = 120$ s, $W_{rev} = 900$ s, and $W_{ret} = 86400$ s. Relations have $\rho = (4, 3, 2, 1)$ for causal/session/endpoint/entity and n_R counts satisfied classes; candidates sort by $(-\rho, -n_R, \Delta_t, t_{start}, id)$. A hub has at least 25 relations in 600 s. Merge permits at most three parents connected by causal, session, or endpoint relations and at most 500 events/200 entities; otherwise attach to the first candidate. The event order is $(t^{event}, t^{ingest}, source_instance, event_id)$.

First finalization freezes t_f and deadline $t_f + 900$ s. The lineage then leaves the open index and remains in a revisable index; at closure, canonical relation keys remain in a closed-lineage index through deadline plus W_{ret} . Each event queries revisable/closed indexes first and returns after exactly one successor/LateEvidenceRecord; only no match permits open start/attach/merge. A qualifying late event before the fixed deadline creates a finalized successor without reopening or extending the horizon; the deadline creates a membership-identical closed successor. A later related event creates only a LateEvidenceRecord in the late store with reason beyond_revision_horizon, closed_lineage_match, or duplicate_after_terminal_outcome. Replay equality covers every state, late record, output, reference, and digest.

A.3 Thresholds and Sufficiency Rules

Table 7 is the single numeric and precedence registry for the submission PDF.

Feature Extractor emits only detected, examined_none, or not_examined. Scope Resolver combines that observation with the

Table 7: Frozen numeric registry and deterministic precedence.

Owner	Fields	Primary value	Freeze / rule
Formation	gap, span, watermark, revision, retention	300/3600/120/900/86400 s	before final train fit
Formation guards	hub/window; merge parents; component	25/600 s; 3; 500 events/200 entities	hard rejection, not learned
Feature	detector IDs; top- k ; sources	3+3 IDs; 3; EDR+NDR	finite [0,1], mid-rank CDF
Label/match	$\theta_{mal}, \theta_{contam}, \theta_{match}$	.50/.25/.50	before label release
Match predicates	time IoU; entity Jaccard	.10/.50	plus exact marker/action
Group gate	positives; negatives; runs; folds	50; 50; 10; 5	deterministic balanced run folds
Gate improvement	Brier; NLL; uncertainty	.01; .01; lower 95% > 0	all folds nonnegative; tie/undefined fallback
Uncertainty	bootstrap repetitions/seed	2000/20260710	complete run clusters
U precedence	coverage	unknown > unavailable > degraded > complete	first matching health/parser predicate
	provenance	missing > partial > complete	required root missing before optional reference
	lateness	late_accepted > revisable > stable	accepted successor; before deadline; closed
	contradiction	present > unevaluable > none	first match over CR1-CR3
	calibration	exact κ enum	out_of_support, supported_group, supported_global

unique covering SourceCoverage interval and emits available, no_detection, collector_unavailable, parser_invalid, or unknown; the first two are health-support states, but zero eligible detector values still force out-of-support. CR1 is an overlapping endpoint-to-host binding conflict, CR2 is canonical EDR/NDR flow-direction conflict, and CR3 is conflicting parent/root lineage for one process key. Health is healthy for zero missed 30 s heartbeats, lag ≤ 120 s, valid parser, and live transport; degraded for one missed heartbeat or lag (120, 600] s; and down for two misses or authenticated stop. Required provenance is every membership/root, selected link, scored input, coverage object, and version; diagnostics are optional. Any later executable threshold file must exactly match Table 7.

A.4 Calibration Resolver and Invalidation

The global fit uniquely minimizes natural-prevalence NLL plus $10^{-6}(a^2 + b^2)$ with $a \geq 10^{-6}$. A registered group key is the RFC 8785/SHA-256 digest of deployment profile and EDR/NDR detector versions; labels, scenario, run, and identity are forbidden. Five-fold calibration-partition cross-fitting keeps each run intact. For each group, runs with binary counts (p_r, n_r) sort by decreasing $p_r + n_r$, then SHA-256 of seed/group/run. Each run is greedily assigned to the fold minimizing the resulting maximum normalized positive, negative, and run loads, then total normalized load and fold ID; the assignment map is sealed and hashed. The gate uses positive improvements $\Delta B = B_{global} - B_{group}$ and $\Delta N = NLL_{global} - NLL_{group}$. It evaluates the gate, after which a passing group is refit on all calibration records for that key and sealed with its coefficients. Manifest validation rejects duplicate keys. Runtime resolution first checks exact mask 11, health, parser, resolved detector states, and formation/feature/model/label/schema/pipeline versions; mismatch yields out-of-support. Otherwise one enabled exact group yields supported_group, and no enabled key falls back to supported_global. Scope Resolver alone emits the key, status, and eligibility; Calibrator only emits P . Episode/candidate, feature/normalizer, raw-model, or label changes invalidate calibration; additive unused fields require equivalence proof.

A.5 Lab, Cards, and Offline Labels

The lab, card, and label contract is exactly Sections 4.2–4.4, Equation 10, and Table 4. It pins CALDERA tag dd2d543..., peeled commit b24f6e7a..., and OCSF 1.8.0; missing manifest fields fail closed; each card/mode requires 10 valid runs with achieved counts pending; and M5-ID remains post-inference only. Each Linux-only M/B card closes image key and digest, ordered ability IDs, no-profile Bash executor plus exact command templates, monotonic offsets, fixed target paths, offline receipt preimage, success/failure predicates, cleanup commands and digest check, and allowed mode. Target commands cannot receive run/card/ability/operation/receipt identifiers or values; orchestrator-only receipts remain in Ground Truth Store. Template variables resolve only from the frozen run manifest. Malicious modes are attack-only or one hash-selected B1–B8 workload begun 60 s earlier; benign cards also run standalone. EB1 uses run-start-anchored 300 s bins; EB2 uses exact shared-entity components within those bins. SB1 takes maximum root-deduplicated finite native severity; SB2 uses the binary SB3 cohort, unconstrained intercept, nonnegative D weight, and the registered L-BFGS-B contract. Missing values and malformed inputs are counted, not zero-filled.

A.6 Populations, Splits, Interventions, Metrics, and RT Schemas

The split is executable in every 10-run card-by-mode stratum: hash ranks 0–4/5–6/7/8–9 map to train/development/calibration/locked-test. Formation coverage, fragmentation, and over-merge use the threshold-qualified graph; one-to-one assignment is restricted to episode F1. Decision counts use C_{dec}^{all} ; scorable coverage uses C_{dec}^{score} ; all class-dependent metrics use positive/negative C_{dec}^{binary} . Outcome and failure labels are joined only after terminal insertion and report supported-group/global, OOS, and failure rates over the all-outcome denominator. The stream population includes both terminal outcome types. RP2–RP7 instantiate every EDR/NDR source by dose cell. RP0–RP7 use the domain-separated HMAC function $U(d, k)$, canonical event/source order, explicit RP3 interval index, exact RP5 buffer width, and byte-level RP7

duplicate-ID preimage specified in Section 4.6. Formal RT4 instantiates 41 children for each of 28 malicious locked-test roots (1,148 bundles); non-locked roots are not read. PO1 is a dedicated locked-test cohort of exactly 40 traces (two sources by two durations by 10 repetitions), balanced over M1/M3/M4/M7; both comparators reuse each immutable trace. It fixes anchor, ready, 1800 s follow-up, and detection/recovery censoring; RT6 worker/repetition measurements are normative as written in Section 4.6. Populations, split, interventions, metrics, and uncertainty are exactly Sections 4.1, 4.6, and 4.7 plus Table 3. Table 6 is the only RT1–RT7 registry. The required supplementary TrustEpisode_reproducibility_companion_v1.zip contains the closed Draft 2020-12 schema, examples, detector/card/baseline/perturbation/result registries, and 17-entry SHA-256 manifest. Its archive SHA-256 is 6298bd2e89bd50a78a890b62750dcca2bcee34906fd6dd73a4a77f2a19c81205; a missing or mismatched upload makes the machine-contract claim unverified.

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